NOTE **Although this performance might be disappointing (after all, we have spent a lot of time building up to this PoI policy that seems overly exploitative), analyzing what is happening will give us insights into how to refine and improve (no pun intended) our performance.**

We note that although PoI gets stuck at the local optimum above, it is doing what it's supposed to do. Specifically, since PoI seeks to improve from the current incumbent, the policy finds that slowly moving to the right of 1 will achieve that with high probability. Although PoI constantly finds more and more improvement by slowly moving to the right, we view this behavior as overly exploitative, as the policy is not exploring other regions enough.

IMPORTANT **In other words, even though what PoI is doing is consistent with what we initially wanted to achieve—namely improving from the incumbent—the resulting behavior isn't what we want. This means that simply pursuing improvement of any kind from the incumbent shouldn't be what we care about.**

There are two ways to fix this overly exploitative behavior. The first is to restrict what we mean by "improvement". Our experiment with PoI shows that at each iteration, the policy only finds marginal improvement from the incumbent by slowly moving in the direction that the GP believes the function moves upward in.

If we were to redefine what we mean by "improvement" by saying that an improvement from the incumbent is only valid if it is at least ε greater than the current incumbent value, and modifying PoI accordingly, then the policy will be more likely to explore the search space more effectively. This is because the GP will know that staying at a local optimum won't lead to significant improvement from the incumbent. Figure 4.14 illustrates this idea.

©Manning Publications Co. To comment go to liveBook
https://livebook.manning.com/#!/book/bayesian-optimization-in-action/discussion

Licensed to ashwin sampath <ashwin.manning@gmail.com>